

Daniel Augusto Muñoz Viveros

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Professional Profile

Data Scientist specializing in artificial intelligence and advanced analytics. Experience in developing and deploying predictive models, applying MLOps and cloud solutions. Focused on scalable solutions for decision-making.

Education

Master of Science in Information Processing

Universidad Autónoma de Zacatecas, Mexico

August 2024

Electronic and Telecommunications Engineering

Universidad del Cauca, Colombia

July 2022

Professional Experience

Research Assistant (Data Science)

Universidad Autónoma de Zacatecas, Mexico

(aug. 2022 – Present)

- Evaluate vaccine coverage, the occurrence of adverse effects, and the immune response following COVID-19 vaccination, applying machine learning techniques to identify significant relationships and support public health decision-making.
- Conducted identification factors that influence antibody levels in post-COVID-19 patients with a public health approach, determining significant clinical risk Factors.
- Analyzed data from the Immunotoxicology Laboratory at the Universidad Autónoma de Zacatecas using Python to develop predictive models, identifying key determinants in post-COVID-19 antibody response.
- Applied MLOps practices in the development of predictive models, using MLflow for experiment tracking, version control, and metric management, enhancing traceability and reproducibility in studies on post-COVID-19 antibody response.

Data Scientist - Freelancer

UpWork (Remote, Mexico)

(jan. 2024 – mar. 2024)

- Validated and corrected SQL queries to ensure data integrity, contributing to the reliability of the analyses performed.
- Developed solutions to export data using ERP APIs, improving data integration and accessibility.
- Organized, transformed and cleaned data, allowing for detailed and consistent analysis in prediction projects.

Jr. Developer

JOMI (Remote, Colombia)

(feb. 2023 – dec. 2023)

- Contributed to the development of multiple web development projects, both front-end and back-end, improving the user experience.
- Responsible for the creation and maintenance of databases and the implementation of business logic in Python, optimizing data management.

Data Scientist - Project Member

DS4A Colombia – Correlation One & MinTic

(mar. 2022 – jul. 2022)

- Developed a weekly sales forecast model using ARIMA with Python, accurately predicting sales peaks to optimize resource allocation and operational planning for a delivery and shipping company in Colombia.
- Performed data cleaning, exploratory data analysis and feature engineering with Python, utilizing Plotly for interactive visualizations to uncover seasonal trends and key correlations.

Undergraduate Researcher

Universidad del Cauca, Colombia

(aug. 2019 – jul. 2022)

- Developed a system using machine learning and computer vision for granulometric analysis of gravel, reducing execution time by nearly 5 times while maintaining precision.
- Led the research project, organizing a work plan, delegating tasks, and ensuring objectives were met on time.
- Adapted to unfamiliar research topics, demonstrating problem-solving skills and quickly mastering new technologies as machine learning and computer vision with Python.

Projects**2026 World Cup Match Predictor**[Visit App](#)

- Developed a predictive system for international football match scores using machine learning models trained on more than 49,000 historical national team records.
- Performed feature engineering based on 17 predictive variables, integrating offensive and defensive metrics, Elo ratings, head-to-head records, and contextual factors to enhance the representation of competitive performance.
- Implemented and deployed a web application to generate real-time predictions, providing an interactive tool for the simulation and analysis of FIFA World Cup 2026 matches and other competitions.

2026 World Cup Prediction Pool Platform[Visit App](#)

- Developed a platform for managing and evaluating predictions related to the FIFA World Cup 2026, automating score calculations and leaderboard generation.
- Implemented data acquisition and processing workflows using external APIs to incorporate up-to-date match information and tournament statistics.
- Developed dynamic visualizations to track match outcomes and participant performance, enabling the monitoring of more than 100 tournament matches.

Multi-class Prediction of Cirrhosis Outcomes[\[GitHub Repository\]](#)

- Developed a predictive model using XGBoost to classify liver cirrhosis outcomes, enhancing performance through advanced hyperparameter tuning.
- Containerized the model with Docker and deployed it on AWS Lambda for scalable and efficient execution.
- Achieved over 90% accuracy, sensitivity, and specificity, validating the model's effectiveness with rigorous performance metrics.

Human Stress Detection in and through Sleep[\[GitHub Repository\]](#)

- Developed a machine learning model to accurately detect human stress levels during sleep, leveraging physiological data for enhanced prediction accuracy.
- Containerized the model using Docker, ensuring a reproducible development environment and streamlined deployment process.
- Deployed the solution on AWS, achieving high scalability and operational efficiency for real-time stress monitoring.

Technical Skills

- **Programming Languages:** Python, R
- **Libraries ML/IA:** Scikit-learn, TensorFlow, Keras, NumPy, Pandas
- **SQL:** PostgreSQL, MySQL
- **Visualization:** Plotly, Matplotlib, PowerBI
- **MLOps y Cloud:** MLflow, Docker, AWS, Prefect

Languages

- Spanish (native)
- English (advanced)

Certifications & Courses

- AWS Certified Cloud Practitioner – Amazon Web Services (2025)
- MLOps: Despliegue de Modelos de Machine Learning – Platzi (2025)
- PowerBI – Santander Open Academy (2025)
- Machine Learning Zoomcamp – DataTalksClub (2024)
- Data Science for All – Correlation One (2022)